# EXECUTIVE SUMMARY

This study aimed to analyze, design, and propose a Business Process Automation plan for the Gonzaga Dental Clinic Management Automation System (GDCMAS). The research focused on streamlining patient records, appointment scheduling, and dental clinic operations across the clinic's two branches.

The clinic currently operates using a paper-based system where patient records are stored in physical envelopes, binders, and cabinets. Appointment schedules are maintained in a logbook, and treatment updates are recorded on index cards. This system presents several challenges including difficulty in locating patient records, lack of accessibility between branches, risk of duplicate records, scheduling conflicts, and the absence of a backup system.

The researchers designed and developed a centralized, web-based system using Next.js 16, React 19, TypeScript, Tailwind CSS 4, and Supabase. The system includes modules for patient management, treatment records with FDI tooth charting, appointment scheduling, consent documentation, public patient booking, staff management, and governance features with comprehensive audit logging.

The system implements a three-layer security approach with role-based access control for Dentist, Secretary, and Administrator. The implementation addresses all identified problems by providing centralized patient records, automated appointment scheduling, digital documentation, secure access control, and a public booking portal.

While some clinic-specific configurations remain pending validation, the core functionality has been successfully implemented. The researchers recommend system validation, staff training, data migration planning, and compliance with the Philippine Data Privacy Act before full deployment.